Unit 2 - Computer Systems
Assignment 6 - Planned Maintenance and Evaluation

Introduction

Scenario/Vocational Context: *You are newly appointed computer technician working for a Jersey Bank. You have been tasked to carry out several installations, upgrades, testing and planned maintenance tasks to demonstrate your skills in this area. You are required to carry out planned maintenance tasks, evaluate and recommend improvements.*

Task 1

(U2:P8) - Maintaining the system

Task description: *Run and record the results (Base Score and sub Base Scores) of the performance monitor to baseline the performance of your system. You will be required to run the program again after you have performed all the maintenance tasks so you can record any changes to the values.*

Company Name: Jersey Bank PLC
Contact: IT Manager
Company Address: 9 Hill Street, St Helier
Company Phone: 00 44 1534 678456

Work Record

Summary of task:

Undertake routine maintenance tasks on a standalone computer system

Assignment Number: 2.6 Technician: Leandro Baroca

Task Number: 3 Criteria Number: 2P8

Environment: (Software): Windows 10

User Platform: (Hardware): x64

Detail of work carried out:

1.Run and record the performance monitor on your computer

Firstly, I opened up the performance monitor by typing “performance monitor” on the search bar/cortana, then I pressed the performance monitor button on the left. I added more counters to the performance monitor by pressing the green plus button. Then I added “%Disk Time” under the physical disk tab and “%Committed bytes in use” under the Memory tab. I checked for any peaks in the memory, CPU or disk performance by looking at the graph and checking how high all the graphs are peaking. There were some very high peaks for all the 3 performance tasks, especially for memory and disk, this means that I will have to maintain the computer to fix the problems.

2.Check the organisation of folders and naming of files, change any inconsistencies

Firstly, I opened the file explorer on windows to check for any clusters of files and folders or unnamed files and unnamed folders. When I clicked on some folders I can see that there are many clusters of unorganised files and unnamed folders which needs addressing. Then I addressed the files by placing them in categorised folders, with the help of unnamed folders, and I also name the files that are unnamed or filled with random characters with appropriate names. Finally, after the files have been organised, I continue checking for any inconsistencies to address. Afterwards, I was confident that the structure of the files and folders were organised and consistent.

3.Backup the documents to an external device

Firstly, I will need to check for the documents which I find are very important to backup to an external drive, basically prioritising documents. Then I will get an external harddrive that is either a standalone hard drive or another PC to backup the documents to. I can also use cloud storage as a backup just incase both the hard drives are corrupt or have any other problems. Finally, I backed up all my documents on an external hard drive and cloud storage so the work wouldn’t be lost.

4.Check how Windows updates are installed (Automatic/Manual)

Firstly, I search for “windows updates” on the search bar/cortana and I click on “Windows Update Settings” to launch the application. On the application, I notice that the windows updates are automatic by default and cannot be changed. However, there is an option to change the active hours for automatic updates, which is the hours in which the PC will not automatically update. After I changed the active hours, I manually check for updates for the PC. Lastly, I saw that the windows OS was up to date and didn't need any new updates.

5.Move any unused documents to an external device

Firstly, I will need to check for the documents which I haven't used in a long time to an external drive, I will need to check for the date last used an look for the ones that were used the longest time ago. Then I will get an external harddrive that is either a standalone hard drive, another PC to backup the documents to or cloud storage as a backup. Finally, I backed up all my documents on an external hard drive and cloud storage so the work wouldn't be lost.

6.Perform a defragmentation on your hard disk

Firstly, I search for "defrag" on the search bar/cortana and I click on "Optimise Drives" to launch the application. On the application, I click on the Windows C: drive so I can defragment the main drive. Then I clicked on the "Optimise" button to launch the defragmentation and I waited for the application to finish the defragmentation process. Finally, after the process was done, the PC would already be defragmented since the application is automated.

7.Run disk Clean-Up

Firstly, I search for "cleanup" on the search bar/cortana and I click on "Disk Cleanup" to launch the application. On the application, I click on all the tick boxes so I can do a full cleanup of the PC. Then I clicked on the "clean up system files" button to launch the cleanup and I waited for the application to finish the cleaning process. Finally, after the process was done, the PC would already be clean from the files since the application is automated.

8.Clean the keyboard and mouse

Firstly, I checked the computer peripherals for any signs of dirt or corrosion by checking all angles and opening them up without damage or modification. I then checked that that there is a lot of dirt within the computer mouse and keyboard. Then, I get a cleaning kit to remove the dirt from the keyboard and mouse. Finally, after the mouse and keyboard has been cleaned, I could see that it was a lot easier to use the buttons and that it also looked a lot better than it previously did.

9.Replacing the printer paper

Firstly, I checked the printer if there are any papers to see if it needs refilling, I also check if there is any paper jams or ripped paper that was left unnoticed. I also checked If there was a lack of paper or torn/jammed paper, just incase I had to remove the torn/jammed paper and I go collect fresh paper for the printer. After checking, I carefully install the paper into the printer while also checking if the new paper is torn and if it is put on the correct location to avoid jamming. Finally, after I installed the paper I turn the printer on and do a printing test to check if the paper prints out correctly without any errors.

10.Replace the damaged external monitor cable (bent pins)

Firstly, I tried to use the external monitor cable, but it did not fit within the port. So, I checked the back of the external monitor cable, noticing that it had bent pins. I immediately turned off all the power to the PC to avoid any electrical hazards when replacing the wire. When the PC powered down, I removed the damaged cable and put it in a place where it would never be used again. Finally, I got a new external monitor cable to replace the old one, after I fitted it in I turned the PC on to check if the cable works.

11.Run and record the performance monitor on your computer

Like previously done before, I open up the performance monitor by typing “performance monitor” on the search bar/cortana, then I press the performance monitor button on the left. I add more counters to the performance monitor by pressing the green plus button. Then I add “%Disk Time” under the physical disk tab and “%Committed bytes in use” under the Memory tab. I check for any peaks in the memory, CPU or disk performance by looking at the graph and checking how high all the graphs are peaking. Finally, I can see that the performance of the computer is much more stable and consistent when compared to the last test.

Task 2

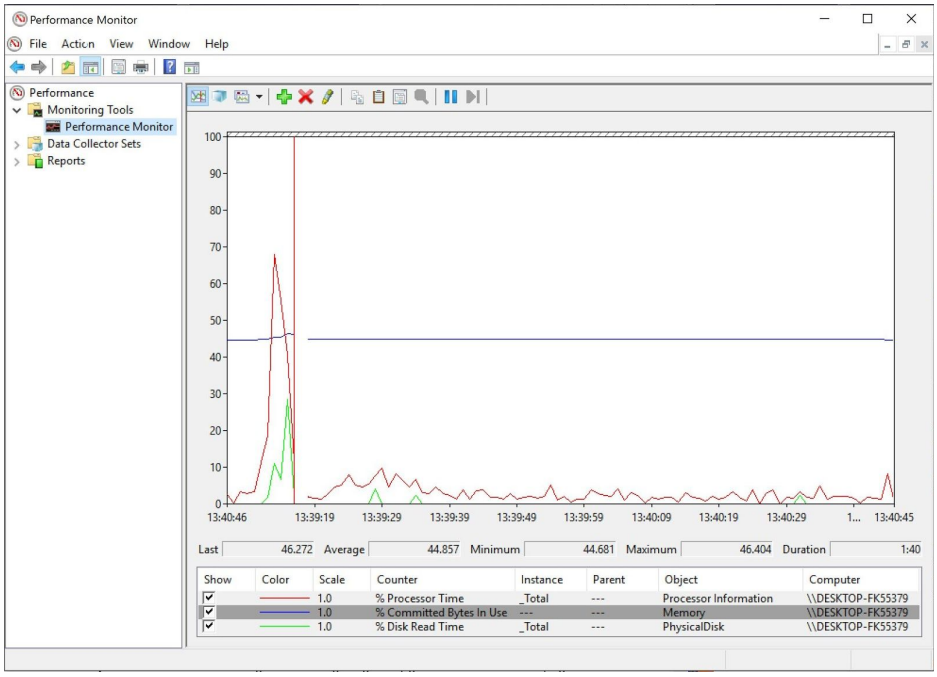
(U2:M3) - Evaluating the system

Task description: *Reviewing the results of the performance monitor and your testing give:*

Introduction

The computer I will be evaluating is my personal laptop. I have tweaked various things to suit my personal needs, such as heavily modifying the user interface with windows and installing third party applications to change the look of my desktop. It is a Microsoft Surface book. It has an Intel Intel Core i5-6300U 2.4GHz dual-core processor with up to 3 GHz with turbo boost and has 8GB RAM. It includes a 256GB PCIe3.0 SSD and Intel HD graphics 520 graphics card.

2 detailed examples of where the system is performing adequately



This screenshot shows the PC's performance with only 2 google chrome windows open in the background. The reason that there is a peak next to the red vertical line is that a software application has been opened, causing it to peak due to the usage it takes.

If a system was to perform adequately, the graph would show that there will be no peaks with any of the tasks that have been added to the performance monitor. To show this the user will need to add the different usage in PC components, whether it is hardware or software, to showcase if it being used heavily or not.

For example,The red line indicating the processor time be very close the the bottom of the graph. Which means that it is carrying out very little tasks which aren't using all the CPU, giving it space for other applications. This essentially means that the processor can handle many different applications at once, giving the ability to multitask.

Another example would be the blue line indicating the committed bytes in use (or the available bytes) would be very close to the bottom of the graph or staying at a very consistent line. This means that there aren't any applications that are using up much of the RAM. If it has a

consistent line, it means that only the windows OS or other background applications are using the RAM at a consistent rate, meaning that there will rarely be any peaks of memory usage.

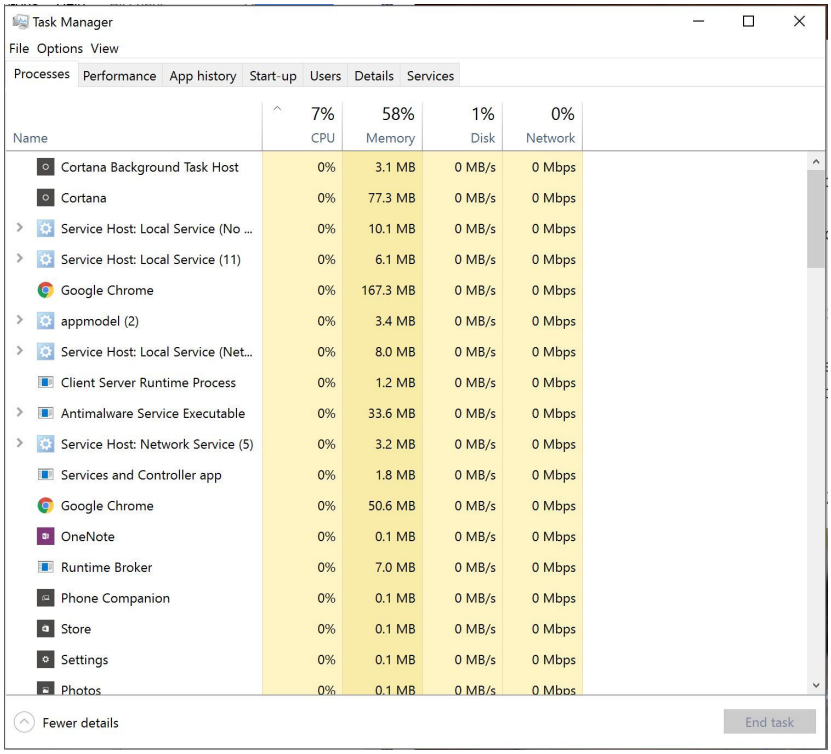
Lastly, the green line indicating disk time would be near the bottom of the graph. This means that the disk is being used to load up any applications or other forms of software, sometimes even the OS can peak the disk usage time.

On the graph, you can see that the disk read time is little to no usage, which means that the disk is performing adequately due to the fact that the disk is barely being used. Which means, as stated before, that the disk is being used to load up any applications or other forms of software, sometimes even the OS can peak the disk usage time.

You can also see that the processor time is being used due to the red lines being visible all the time. However, the usage is miniscule. Which, as I stated before, means that it is carrying out very little tasks which aren’t using all the CPU, giving it space for other applications.

With the committed bytes in use for the memory, the line is very consistent. Which means that the memory is at constant use, but it isn't at its very limit, so it is performing at a very good rate. Which means that, as I once again stated above, only the windows OS or other background applications are using the RAM at a consistent rate, meaning that there will rarely be any peaks of memory usage.

This graph suggests that the processor, memory and disk can multitask very well when the applications are already loaded up and idle, meaning that the user can have many applications open in the background when operating this computer system.



This screenshot is a screenshot of the task manager. The task manager basically shows the different applications that are open in the background on the processes tab, which is the tab that is open on the screenshot.

If a system was to perform adequately, the table would show that the numbers which is indicated by each application would be a small number or a 0.

For example,The column showing the CPU and the percentage will be at a very low number, such as 7%. Which means, just like the performance monitor, that it is carrying out very little tasks which aren't using all the CPU, giving it space for other applications. This essentially means that the processor can handle many different applications at once, giving the ability to multitask.

You can also see that on the table, just like the performance manager, the processor time is being used due to the 7% of total usage. However, the usage is miniscule. Which, as I stated before, means that it is carrying out very little tasks which aren't using all the CPU, giving it space for other applications.

Another example would be the memory column which is showing 58% usage, even though a screenshot cannot show it, the number stays consistent just like the performance monitor. This means, just like the performance monitor, that there aren't any applications that are using up much of the RAM.

For the memory, just like the performance manager, the percentage is at a consistant 58% usage. Which means that the memory is at constant use, but it isn't at its very limit, so it is performing at a very good rate. Which means that, as I once again stated above, only the windows OS or other background applications are using the RAM at a consistent rate, meaning that there will rarely be any peaks of memory usage.

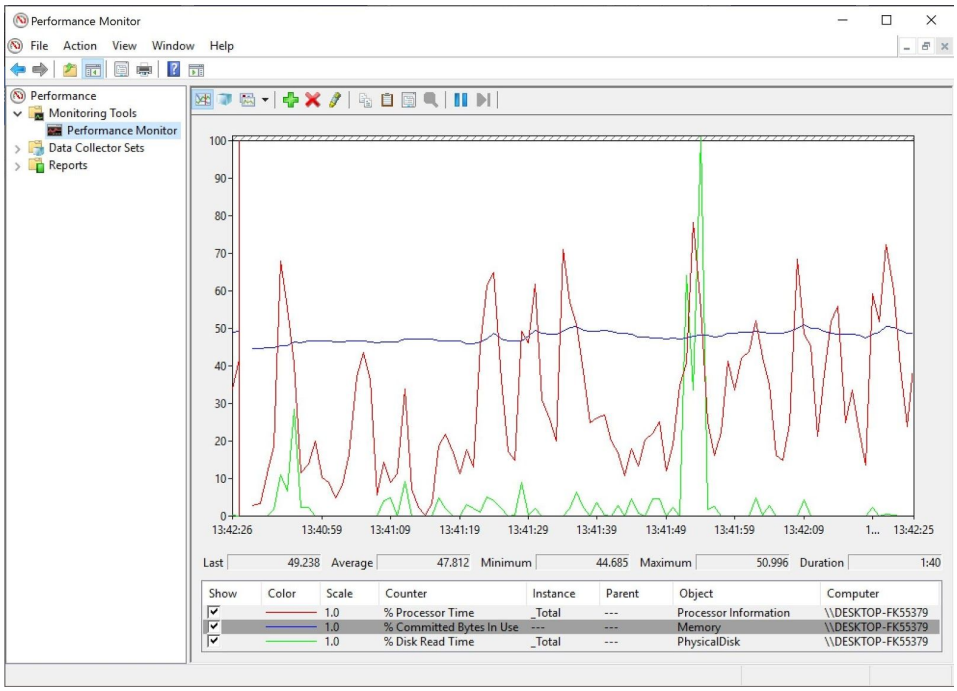
Yet another example is the disk which is indicated at 1%. This means that the disk is being used to load up any applications or other forms of software, but it is a very miniscule amount. This means that the user can load up a lot of applications without the disk being affected heavily.

On the table, just like the performance manager, you can see that the disk is little to no usage, which means that the disk is performing adequately due to the fact that the disk is barely being used. Which means, as stated before, that the disk is being used to load up any applications or other forms of software, sometimes even the OS can peak the disk usage time.

Lastly, the network column shows that it is at 0%. This means that no network is being used to load up any applications that require internet access.

This table suggests that the processor, memory and disk can multitask very well with the many idle applications are already loaded up, meaning that the user can have many applications open in the background when operating this computer system.

2 detailed examples of where the system is underperforming



This screenshot shows the PC's performance when I open many applications over an extended period of time. As you can see, there are many peaks with the processor time, committed bytes in use for the memory and the disk read time.

If a system was underperforming, the graph would show that there will be a lot of peaks with the tasks that have been added to the performance monitor.

For example,The red line indicating the processor time be peaking to the top of the graph. Which means that it is taking a long time carrying out the tasks. This essentially means that the processor is handling many different applications at once.

Another example would be the blue line indicating the committed bytes in use (or the available bytes) would be peaking to the top of the graph. This means that there are many different applications that are using up a lot of the RAM.

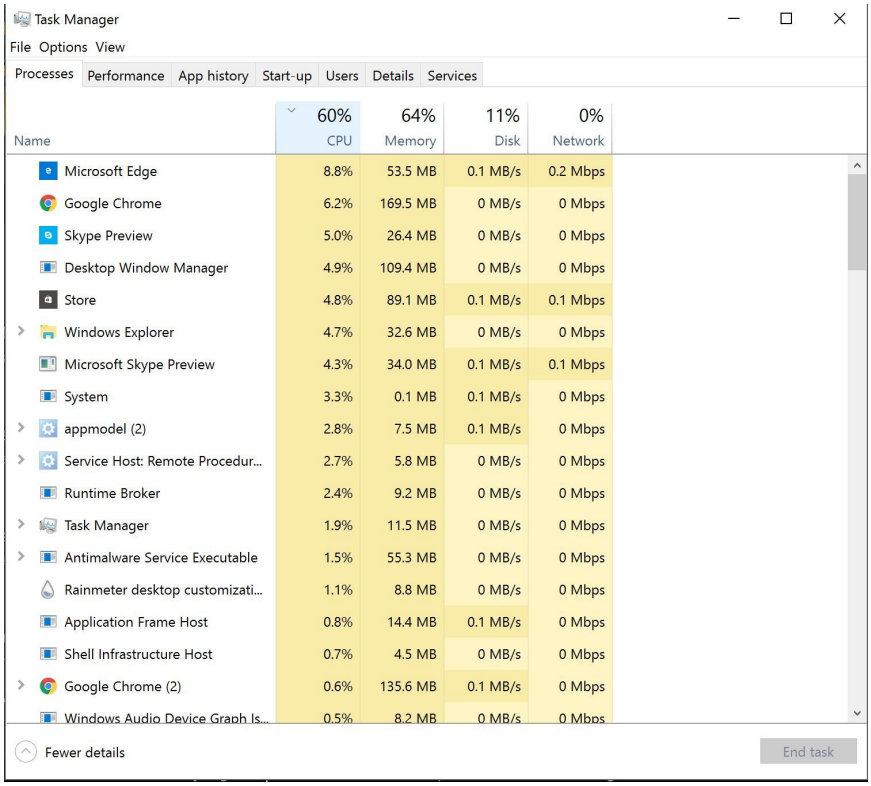
Lastly, the green line indicating disk time would peak to the top of the graph. This means that the disk is being used a lot to load up any applications or other forms of software, which is taking the disk usage to the limit.

On the graph, you can see that the disk read time is little to no usage, which means that the disk is performing adequately due to the fact that the disk is barely being used. Which means, as stated before, that the disk is being used to load up any applications or other forms of software, sometimes even the OS can peak the disk usage time.

You can also see that the processor time is being used due to the red lines being visible all the time. However, the usage is miniscule. Which means that it is carrying out very little tasks which aren't using all the CPU, giving it space for other applications.

With the committed bytes in use for the memory, the line is more inconsistent when compared to a well maintained PC. This means that the memory is being used at different rated by different applications,which can put it at its very limit, causing it to underperform.

This graph suggests that the processor, memory and disk are being used a lot when a lot of different applications are being loaded up at once, meaning that having to open to many applications when operating this computer system would cause the PC to reach its limit, meaning that the user will have to wait for all the applications to load for the PC to perform better.



This screenshot is a screenshot of the task manager. In this screenshot, you can see that many applications are being loaded up and that it is using the main components of a PC to load them up.

If a system was underperforming, the table would show that the numbers which is indicated by each application would be a very large number, or in extreme cases 100% usage.

For example, the column showing the CPU and the percentage will be at a very high number, such as 60%. Which means, just like the performance monitor, that it is carrying out a lot of tasks which is using most of the CPU, giving it very little space for other applications. This essentially means that the processor is already handling many different applications at once, reducing the ability to multitask.

You can also see that on the table, just like the performance manager, the processor time is being used a lot due to the 60% of total usage. Which means that it is carrying out a variety of different tasks which is using a lot of the CPU, reducing the space for other applications significantly.

Another example would be the memory column which is showing 64% usage, even though a screenshot cannot show it, the number isn't staying at a very consistent rate. This means, just like the performance monitor, that there are a lot of applications that are using up a lot of the RAM.

With memory, just like the performance manager, the percentage is at an inconsistent usage with 64% average usage. Which means that the memory isn't at a constant use when compared to RAM that is performing well, but it still isn't at its very limit, so it is performing at a somewhat average rate.

Yet another example is the disk which is indicated at 11%. This means that the disk is being used to load up any applications or other forms of software, but it still is a small amount, but it's a lot larger when compared to the adequate performance. This means that the user can still load up a lot of applications without the disk being affected heavily, but not to the extent of the adequate disk performance.

On the table, just like the performance manager, you can see that the disk still has little usage, which means that the disk is still performing somewhat adequately due to the fact that the disk is barely being used.

Lastly, the network column still shows that it is at 0%. This still means that no network is being used to load up any applications that require internet access.

This table suggests that the processor, memory and disk can still multitask very well with the many idle applications are already loaded up. However, if the user decides to load up a lot of applications at once, the processor would be the first component to be put to the limit, with the disk being last due to the disk being an SSD. Overall, the PC does a very good job at multitasking applications, even when they are being launched.

2 examples with reasons of an improvement in performance due to the maintenance tasks performed

With the maintenance tasks that have been performed, the PC has been improved a decent amount when compared to before. All the tasks that have been carried out proved to work due to their ability to make the PC run at a much smoother rate.

These tasks consist of: Check the organisation of folders and naming of files, change any inconsistencies, Backup the documents to an external device, Check how Windows updates are installed (Automatic/Manual), Move any unused documents to an external device, Perform a defragmentation on your hard disk, Run disk Clean-Up, Clean the keyboard and mouse, Replacing the printer paper, Replace the damaged external monitor cable (bent pins) and Run and record the performance monitor on your computer.

The most significant impact of these maintenance tasks is the: Defragmentation and Cleanup.

With defragmentation, some file format systems (such as NTFS) are usually very messy at managing files. When a user saves a new file, it is randomly stored in an empty space within the hard drive. If the user keeps saving files into the empty spaces on the disk, the more the disk

will be scattered or fragmented around the disk due to the files filling up the empty spaces that were left available. This means that it takes more time for the PC to find specific files, slowing down the performance of the PC.

Defragmenting is basically a process of making these files easier to access, especially with a lot of data stored. However, this will only benefit HDDs, which are hard disk drives or magnetic disks, and not SSD.

The reason for this is due to the SSD being a very fast flash storage hardware, it would be obsolete to defragment it because it will make an extremely miniscule difference which wouldn’t make a noticeable performance boost.

Overall, if the PC has a HDD filled with a lot of data, it is recommended to defrag the PC if it has never been done or hasn't been done recently. However, if the PC has a SSD or a HDD with very little data on it, it is not recommended to defrag the PC as it would make little to no performance boost and would be a waste of valuable time due to the extensive amount of time that the defragmentation process takes.

With disk cleanup, over time computers tend to get slower due to the fact unnecessary software is installed and files that have become disorganised and take up a lot of storage space.

One of the many ways to enhance the performance of a computer and free up some space is by using the disk cleanup tool. This is by uninstalling or removing unused programs, software and files. By removing these files this has a positive impact on the computer as it means there is more space on the hard drive.

Also there are a lot of programs on the PC that the user might not use that run in the background of the computer. By removing the programs, it can have a positive effect on the performance of a computer.

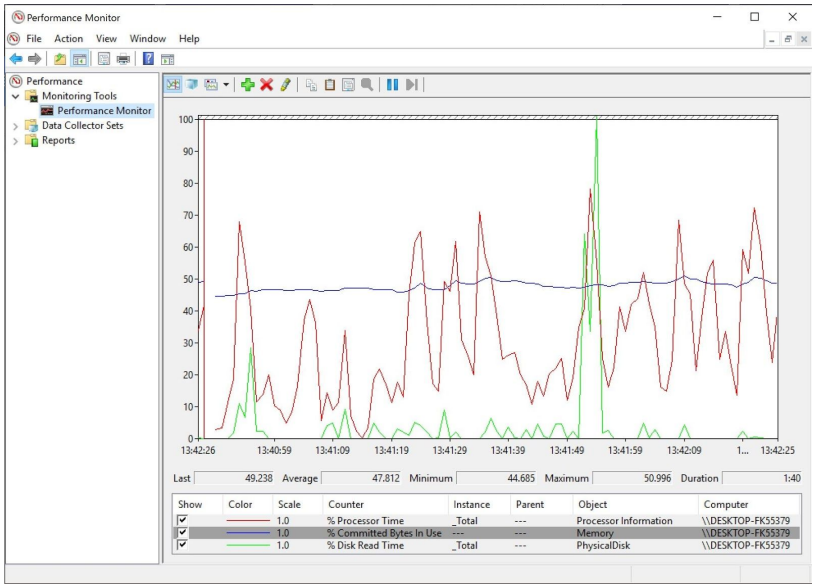
When using a computer, it's easy to have a computer get loaded up with junk that slows it down. The worst part is, this type of garbage is installed into the PC by installing "helpful" utilities, toolbars, and other add-ons.

There are a significant amount of “computer-stalling junk” which can significantly impact the performance of a PC. However, there are some important types of junk that can really impact the performance of the PC, so it would be best to remove them for best performance.

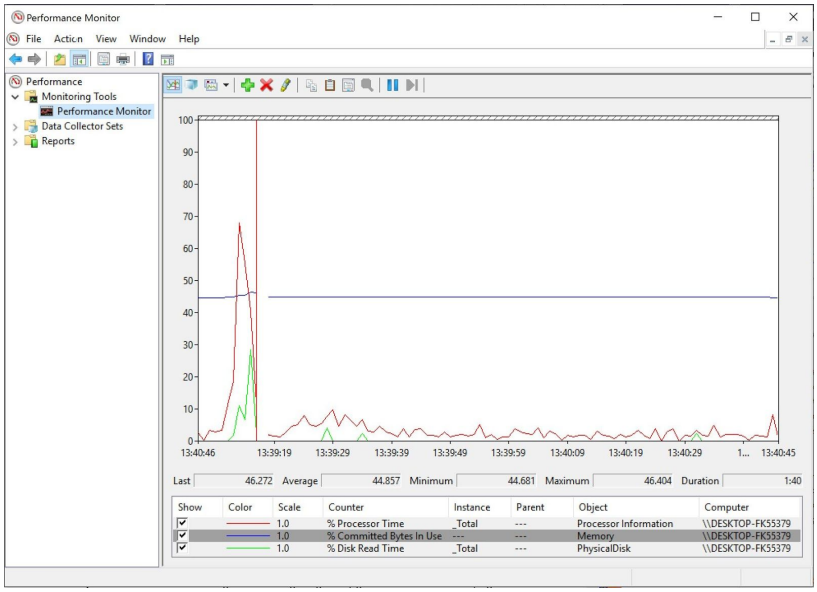
These types of junk can consist of: Automatic update systems for various applications (but be careful: some apps, like Flash, Acrobat, QuickTime, and Web browsers are prime malware targets and you will want to keep these up-to-date).

Other forms of junk that can slow down a PC can consist of: Things that run on startup, Windows services you don't really need, software from the PC maker, Toolbars, Browser, plug-ins (the Skype browser plug-in is an especially bad offender), P2P applications, Web servers and database servers that were installed by since-removed applications.

Below is a screenshot of the performance of the PC before the maintenance tasks.



Below is a screenshot of the performance of the PC after the maintenance tasks.



Overall, it is recommended to use disk cleanup at least once a month and over time it will be noticeable that a gradual improvement in the way the computer is running. By uninstalling and deleting used software and application space is being created on the hard drive resulting in the overall performance increasing.

If there is less data on the hard drive they system dent have many application running at the same time meaning there are fewer thing to slow the device down.

Recommend the specification of a suitable upgrade to the hardware

With a computer, there are some key hardware components that can really extend the life of older hardware, provide a bump in performance and cost much less than the cost of an iPad and its accessories. For example, hardware such as the video card, RAM and the Hard drive (especially an SSD) help improve the performance.

For the video card, many older PCs use a video card onboard the system's motherboard to drive the display(s) attached to the PC. There are a lot of video card options available for PCI and PCIe with 1GB and 2GB of memory that might give the user a higher resolution image than the user is getting right with the current video card.

One thing to keep an eye on when upgrading video cards is that some of the newer ones require additional power and need to plug into the system's power supply or a power port on the motherboard. It would be best to check with a hardware OEM to make sure the system does support that type of option for a new video card.

For the RAM, this memory is used as a temporary storage area of data as it moves to and from your hard drive for processing. Memory prices have fluctuated a lot of the last few years but there are many deals to be found on online sites such as Amazon, NewEgg, TigerDirect and even Best Buy's Marketplace.

Always shop for the best deal and buy as much memory as the budget allows for what the motherboard will support. It is important to check with the motherboard manufacturer or the OEM to know what the PC's memory configuration is exactly.

While some retail stores like Best Buy and others carry RAM on the shelves it tends to be more expensive so it would be best to avoid that and shop online with a reputable retailer with good electronics return policies.

For a hard drive, preferably an SSD, the prices on SSDs have dropped over the last several years, they have become very solid alternatives to the spinning hard disks that are in most machines that are five years old.

Spinning hard drives came in two speeds 5400 and 7200 RPMs. While the 7200RPM drives were faster than the slower 5400 RPM drives, SSDs put those devices way back in performance. Even large SSDs in the 240GB and higher range have come well down in price. A quick search of Amazon for "240GB SSD" shows five results with each priced under $70. Which makes it very easy to obtain two at that price for spare storage as well in the current PC.

Most consumer SSDs use SATA based connections so it's best to make sure that the older PC supports a SATA connection although a large majority of PCs built five years ago were already starting to support that configuration. If there is still a computer based on IDE hard drives, it would be best to use an IDE to SATA adapter to upgrade to an SSD and they cost less than $5 on Amazon.

Recommend the specification of a suitable upgrade to the software

Just like hardware, there are some key software components that can really provide a bump in performance and with the price being a lot less than the hardware upgrades, sometimes the software upgrades are completely free. For example, software such as Windows updates, driver updates and BIOS updates help improve the performance of a computer.

With Windows updates, it can bring security patches that will sacrifice some speed to increase the safety. However, sometimes software updates can bring performance improvements. In reality, the reason that most PC updates either improve or don't improve performance are usually ignored or otherwise postponed.

The reason that windows updates improve the performance of the PC is that the improved security that comes with the update ensures that the PC it is installed in is protected, which means that any external viruses or other harmful forms of software doesn't infiltrate the PC and significantly decrease the performance of the PC.

Overall, windows updates usually don't improve the performance of the PC. However, it does prevent the PC from being impacted from harmful software which will reduce the performance of the PC, but with the a small performance sacrifice for the security.

So, if the hardware on the PC is outdated and if the computer is already running slow, it would be best to be wary about the windows updates that windows released to be installed onto the computer.

With driver updates, this is another important aspect that might actually improve the overall experience on the computer, because an outdated drivers is almost never a good thing. However, drivers usually don't improve performance as significant to newer drivers with newer video cards.

Usually, it depends on the software and the fixes the drivers specifically target. A lot of recent graphics cards patches have been made to improve performance due to new graphics cards and newer software being released.

However, it's very rare to get a large or sometimes even noticeable improvement of any kind from a driver. Mostly, they're fixing issues with individual features. Cards differ enough that their

drivers can make or break whether the PC can run certain software, but hardly ever whether the software will be vastly improved.

Overall, driver updates mostly affect newer cards and newer software. If the software or video card is working as intended then it wouldn't be necessary to update, but using the latest drivers is still the best way to ensure that the user won't have any issues when new software is installed. Basically, it's typically a good idea to update drivers but not always. Some drivers can hurt performance and have bugs.

With BIOS updates, it can improve the performance of the PC when handling software, especially heavy duty software. However, it is not certain what goes into a given BIOS update.

Sometimes BIOS updates contain stability enhancements, security updates, performance enhancements, support for new hardware and even CPU microcode which can further improve performance.

It is recommended to keep the BIOS updated. However, if the user of the PC has messed with the software of the computer, such as overclocking, then the BIOS update will remove the overclock and will need to be re-applied in order to restore the overclock.

BIOS updates in the modern day are much easier. What I mean by this is that most board will allow the user to either flash it from Windows, or put the update on a flash drive and update it from a built-in BIOS updated. However, it is critically important that the machine doesn't get powered down while it's being updated.

Overall, manufacturers are constantly working on the BIOS and most of them have lots of tweaks, fixes and updates. Often, it is an update for the DRAM alone, but it can have other tweaks for drivers, GPUs and other newer components.

Summary of Evaluation

In conclusion, the system has a decent performance. This is mainly because it has new hardware components. The computer is relatively fast and most the time it doesn't struggle to load applications and webpages even if they require a lot of high performance graphics.

When the PC handles multiple applications, the system has good performance providing that they do not require much graphical processing, but the performance can be slown down if many different applications were opened at once. Overall, the computer can cope with a wide variety of software applications and allows the user to use the system quite effectively.

Task 3

(U29:D1) - Justifying an upgrade

Task description: *The Finance Manager is investigating an upgrade to the software (e.g. Windows 7 to Windows 8.1 or Windows 8.1 to Windows 10) and they need the budget to be approved.They would like you to justify the upgrade. The justification should include a business case e.g. balance of costs against perceived benefits e.g. software costs, training `costs, benefits of accessibility and productivity etc. The costs and benefits are to be described in detail and an estimate of their value included. A summary of whether it is cost effective should also be included.*

Task 5 Planning Grid	**Description and detailed example of the cost/benefit**
Description of Software upgrade or installation	This table is going to show the costs and benefits for the upgrade of the operating systems which consist of Windows 8.1 being upgraded to Windows 10.
Costs	In this section, I will be explaining the costs on upgrading the operating system from windows 8.1 to windows 10.
Software	With the software, there are many different costs depending on what type of software that is going to be used. However, all these costs will all be related to the software licenses. With the price of Windows 10 reducing, due to the OS already getting older, it means that it is a lot more affordable to purchase the licenses, especially for a company planning on installing Windows 10 on many different computer systems. The current price of the Windows 10 Professional 64-bit Operating System is £79.95 for one license. This may seem very cheap, but for a business the price can be easily built up, whether it is a small or large business. For a large business, there will be a lot of employees which will be using computer systems due to companies relying on them for the modern world. A large business may need to invest in tens of thousands, maybe even hundreds of thousands of pounds to purchase the licenses for each computer system.

	For a small business, the price won't be as large when compared to a large business. However, due to the business being small, it may still be a very big price for a license for each computer system. A small business would typically buy hundreds or even thousands of pounds worth licenses for each employee that uses a computer system.
Technical staff	Not only will the price of obtaining the software be costly to the company, but the process of the installation of the OS will also be costly. For instance, the time and preparation for the installation of the OS means that the company will have to postpone certain areas of the business in order to install the OS on all the computer systems. This means that the business will be losing out on making revenue from the employees working in the computer systems, increasing the overall revenue cost of installing the updates. However, this can be somewhat countered by the technician installing the updates when the employees and the business is not on a working hour or day.
Training	With the training to use the new software, there are many different costs depending on what type of software that is going to be used. However, all these costs will all be related to the users and technicians for the software. With Windows 10 being a common Operating System for home use, there is a chance that many employees will already know how to operate the Windows 10 OS. However, there are still a few employees which don’t know how to operate Windows 10. This can be due to the fact that they either use a mac/linux pc or that they only used the computer that is on the business. For a large business, there will be a lot of employees which will be using computer systems

	due to companies relying on them for the modern world. A large business may need to invest in thousands, maybe even tens of thousands of pounds to fund the training for the employees. For a small business, the price won't be as large when compared to a large business. Also, it makes it a lot more affordable to train the employees to use the new OS. A small business would typically spend very little money. At most, hundreds of pounds worth of training for each employee that uses a computer system.
Etc.	There are other areas in which the cost can be affected to a business. Such as the managers time, backup facilities and the accommodation. The reason that the managers time will be affected is due to the fact that the manager will be busy on looking after the installation of the Operating Software on the computer systems. Meaning that the manager will be too busy on other, more profitable areas for a business, causing the business to lose out on potential revenue. The reason that backup facilities use a lot of cost is due to the fact that the company will have to invest on the extra facilities which contains the backup for the windows OS. However, this investment can be worthwhile in the long run due to the software being prone of having errors or being corrupt. Which, without a backup, would be more impactful than a backup facility. The reason that accommodation would be costly is due to the fact that the company will have to invest on the extra accommodation which contains the newer computer systems with the new OS. This can be worthwhile in the long run due to the fact that the new accommodation for the new

	computer systems can bring a lot more workforce to the business, hopefully creating enough revenue to surpass the costs.
Summary of costs	In conclusion, the costs for the software, technical staff, training and other reasons prove to be very costly for a large business and a small business. However, that doesn't mean that the costs won't be compensated with revenue that the new software will bring to the company. The new operating system proves to be much faster and contains more features when compared to older operating system. The older systems may be able to carry out tasks well, but the new operating system improves the multitasking and performance of which these tasks can be carried out. Overall, the software upgrade can be worthwhile if all goes well. However, it is still risky to get the upgrade due to the errors that can occur or the fact that the costs won't be compensated.
Benefits	In this section, I will be explaining the benefits of upgrading the operating system from windows 8.1 to windows 10.
Image of Organisation	Getting a newer operating system makes the company look like a much more forward thinking company, which can be beneficial to newer employees and customers. For instance, with an older Operating System, it makes the company look like an older and less modern company when compared to other companies that are much more modern. This may not seem very important at first glance, but with the customers usually going for the better quality and more modern businesses, this can prove worthwhile. For example, if there is a company which still uses windows 7 and another uses windows 10.

	The one with windows 7 may be more comfortable with its current employees and operating system, but newer employees may not like the use of windows 7 due to the fact that they are used to using windows 10. While the company with windows 10 will be much more attractive to newer employees due to the familiar operating system that the company has.
Productivity	The newer operating system will be a lot more productive towards the company when compared to the older operating systems. This means that the company can have much more integration of solutions due to the newer OS. For instance, with an older Operating System, it can still be productive, but due to the outdated software and systems, it cannot match the performance of the newer Operating System. This may not seem important for an employee or manager who just looks forward to getting work done, but with a faster and more advanced operating system, it means that it is a lot more sophisticated and gives a better ability to multitask. For example, if there are computer systems which still uses windows 7 and other computer systems which uses windows 10. The one with windows 7 may be more comfortable with its current employees and operating system, but the performance of the windows 7 OS won't be as good with performance when compared to windows 10. While the company with windows 10 will be a lot better performance wise, which means that multitasking and software launching is a lot quicker, which means more work can be done in a shorter amount of time.
Reliability	The newer operating system will also be much more reliable than the older operating systems. This means that the company can have much more stable platforms and the latest support, which can

	be very cost effective. For instance, with better platforms to produce work in, the risks of the software crashing or having any errors will be minimized due to the more improved and updated software of the Windows 10 Operating System. For example, if there is a company which still uses windows 7 and another uses windows 10. The one with windows 7 may be more comfortable with its current employees and operating system, but the reliability of the windows 7 OS won't be as good as windows 10. This means that there is a higher chance of the software crashing in Windows 7, making it possible that the work can be removed instantly. While the company with windows 10 will be a lot better with reliability, which means that the software has a small chance of crashing and losing work. So there is a smaller chance that the work will be lost due to the errors that are more prone in windows 7.
Etc.	The newer Operating System can improve the recruitment of newer employees for the same reason that it improves the image of the organisation. For instance, with an older Operating System, it attracts a lot more newer employees which have a chance of having good skill that can benefit the company. For example, if there is a company which still uses windows 7 and another uses windows 10. The one with windows 7, just like the image of the organisation, may be more comfortable with its current employees and operating system, but newer employees may not like the use of windows 7 due to the fact that they are used to using windows 10. While the company with windows 10 just like the image of the organisation, will be much more attractive to newer employees due to the familiar

	operating system that the company has. Which can bring much more newer, fresher employees that can bring new skills into the business.
Summary of Benefits	In conclusion, image of the organisation, productivity, reliability and other reasons prove to be very beneficial for a business. However, that doesn't mean that it is guaranteed that it will be as beneficial as it possibly can be. With the new operating system, it proves to attract a lot more newer, fresher employees to the company due to the modern and common aspect of the Windows 10 OS, which can be very beneficial. With the older systems, they may be able to carry out tasks well, but the new operating system improves the image and ability to be more productive and reliable when compared to Windows 7. Overall, the software upgrade can be worthwhile if all goes well. However, it is still risky to get the upgrade due to the risk of the benefits not achieving its full benefits.

Summary of the Cost/Benefit analysis including alternatives (referencing the detail above)

In conclusion, The costs and benefits of getting Windows 10 does have its possibility of improving the company a lot more when compared to Windows 7. However, it's not guaranteed that the possibility may come true. Basically, getting Windows 10 installed in a business, whether it's a large or small business, is risky due to the risks that are still there when installing the new OS.

However, this is a lot more effective on a small business when compared to a larger one. The reason for this is due to a larger business only being dented slightly if it has the risks that are possible with the installation, while a smaller business is risking bankruptcy when it has the risks involved with the installation of Windows 10.

Overall, if it is a small business, it wouldn’t be worth getting Windows 10 due to the fact that it is too risky and won't be as beneficial when compared to a large business. If it is a large business, it would be worth getting Windows 10 due to the risks not being very impactful and that it can benefit the company in the long run, due to the multiplication of the revenue income that the more performance, multitasking, reliability and better image for the company to newer employees it brings.